

Experiment 14: GitHub Collaboration using Fork and Pull Request Workflow

Aim

To demonstrate collaborative software development using Git and GitHub by following the Fork and Pull Request workflow.

Tools Required

- Git
- GitHub Account
- VS Code (or any code editor)

Repository Used

Repository: <https://github.com/Santhosh-1307/Test-repo>

Procedure

Step 1: Fork the Repository

- Open the public GitHub repository.
- Click **Fork** to create a copy under your GitHub account.

Screenshot 1: Forked repository page.

Create a new fork

A fork is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project.

Required fields are marked with an asterisk (*).

Owner * Repository name *

Sriarevalagan23 Test-repo-fork

Test-repo-fork is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

0 / 350 characters

☒ Copy the main branch only

Contribute back to Santhosh-1307/Test-repo by adding your own branch. [Learn more.](#)

☐ You are creating a fork in your personal account.

Create fork

Test-repo-fork Public

forked from [Santhosh-1307/Test-repo](#)

main 1 Branch 0 Tags

Go to file

Code

This branch is up to date with [Santhosh-1307/Test-repo:main](#).

Contribute Sync fork

Sriarevalagan23	Initial commit	12de6f2 · 5 minutes ago	1 Commit
.claude	Initial commit	5 minutes ago	
.vscode	Initial commit	5 minutes ago	
assets	Initial commit	5 minutes ago	
scripts	Initial commit	5 minutes ago	
src	Initial commit	5 minutes ago	
.gitignore	Initial commit	5 minutes ago	
AGENTS.md	Initial commit	5 minutes ago	
CLAUDE.md	Initial commit	5 minutes ago	

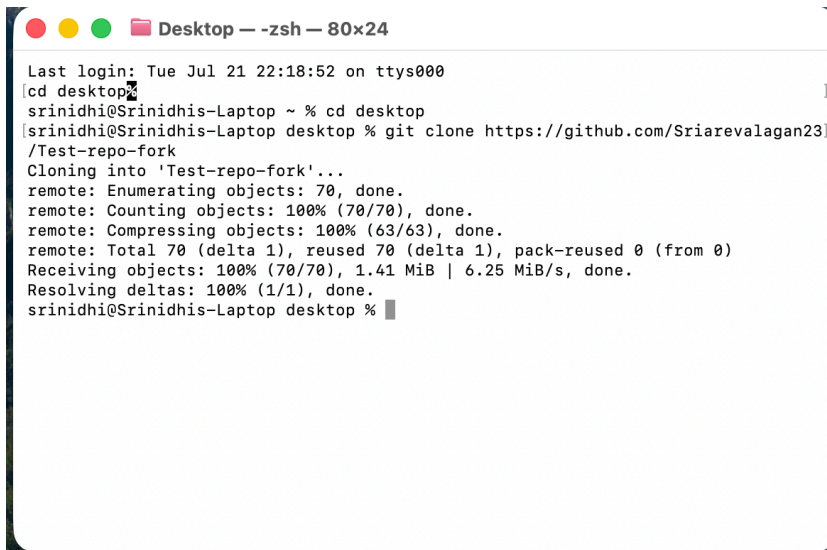
Step 2: Clone the Repository

Bash

```
git clone https://github.com/your-username/repository-name.git
```

```
cd repository-name
```

Screenshot 2: Terminal showing successful clone.



```
Desktop — zsh — 80x24

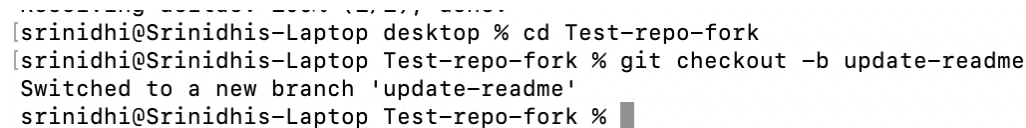
Last login: Tue Jul 21 22:18:52 on ttys000
[cd desktop]
srinidhi@Srinidhis-Laptop ~ % cd desktop
[srinidhi@Srinidhis-Laptop desktop % git clone https://github.com/Sriarevalagan23/Test-repo-fork
Cloning into 'Test-repo-fork'...
remote: Enumerating objects: 70, done.
remote: Counting objects: 100% (70/70), done.
remote: Compressing objects: 100% (63/63), done.
remote: Total 70 (delta 1), reused 70 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (70/70), 1.41 MiB | 6.25 MiB/s, done.
Resolving deltas: 100% (1/1), done.
srinidhi@Srinidhis-Laptop desktop %
```

Step 3: Create a New Branch

Bash

git checkout -b update-readme

Screenshot 3: Terminal showing branch creation.



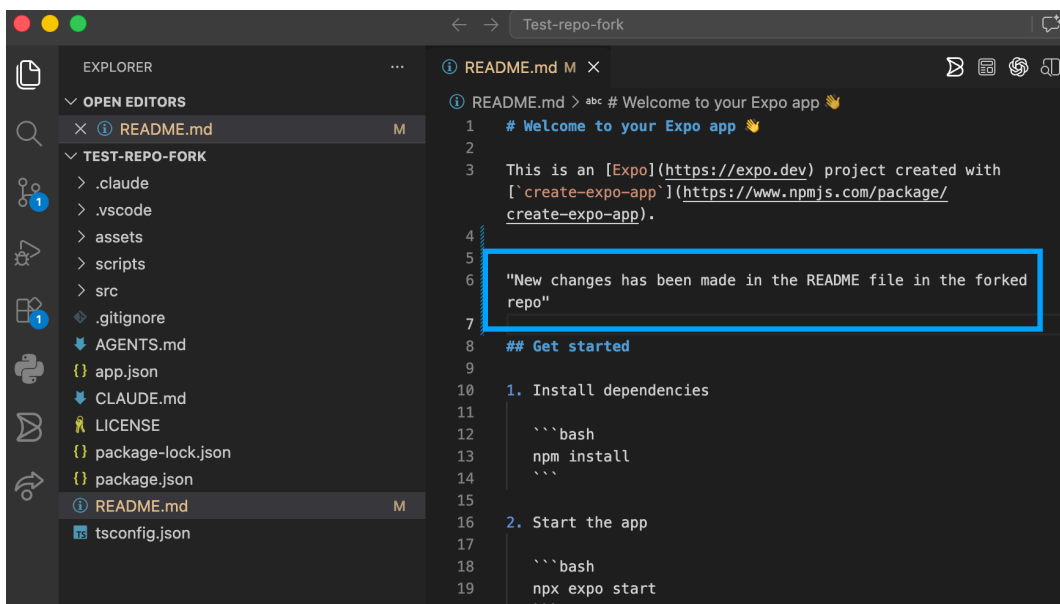
```
[srinidhi@Srinidhis-Laptop desktop % cd Test-repo-fork
[srinidhi@Srinidhis-Laptop Test-repo-fork % git checkout -b update-readme
Switched to a new branch 'update-readme'
srinidhi@Srinidhis-Laptop Test-repo-fork %
```

Step 4: Make Changes

Example:

- Updated README.md
- Added project description.

Screenshot 4: VS Code showing the modified file.



```
Test-repo-fork

EXPLORER
  OPEN EDITORS
    README.md M
  TEST-REPO-FORK
    .claude
    .vscode
    assets
    scripts
    src
    .gitignore
    AGENTS.md
    app.json
    CLAUDE.md
    LICENSE
    package-lock.json
    package.json
    README.md M
    tsconfig.json

README.md M
1 # Welcome to your Expo app 🎉
2
3 This is an [Expo](https://expo.dev) project created with
4 ['create-expo-app'](https://www.npmjs.com/package/
5 create-expo-app).
6 "New changes has been made in the README file in the forked
7 repo"
8 ## Get started
9
10 1. Install dependencies
11
12 ```bash
13 npm install
14 ```
15
16 2. Start the app
17
18 ```bash
19 npx expo start
20 ```
```

Step 5: Commit and Push Changes

Bash

```
git add .
```

```
git commit -m "Updated README with project description"
```

```
git push origin update-readme
```

Screenshot 5: Terminal showing successful push.

```

srinidhi@Srinidhis-Laptop Test-repo-fork % git add .
srinidhi@Srinidhis-Laptop Test-repo-fork % git commit -m "Updated README with project description"
[update-readme 5e58284] Updated README with project description
 1 file changed, 3 insertions(+)
srinidhi@Srinidhis-Laptop Test-repo-fork % git push origin update-readme
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 385 bytes | 385.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'update-readme' on GitHub by visiting:
remote:   https://github.com/Sriarevalagan23/Test-repo-fork/pull/new/update-readme
remote:
To https://github.com/Sriarevalagan23/Test-repo-fork
 * [new branch]      update-readme -> update-readme
srinidhi@Srinidhis-Laptop Test-repo-fork %

```

Step 6: Create Pull Request

- Open GitHub.
- Click **Compare & Pull Request**.
- Enter title and description.
- Click **Create Pull Request**.

Screenshot 6: Pull Request page.

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons here](#).

 base repository: Santhosh-1307/Test-repo  base: main   head repository: Sriarevalagan23/Test-repo-f...  compare: update-readme 

 **Able to merge.** These branches can be automatically merged.

 **Add a title ***
Updated README with project description 

Add a description

Write Preview

new changes have been made in the read me file.

 Markdown is supported  Paste, drop, or click to add files

☒ Allow edits by maintainers 

Create pull request 

Reviewers
No reviews 

Assignees
No one—assign yourself 

Labels
None yet 

Projects
None yet 

Milestone
No milestone 

Development
Use [Closing keywords](#) in the description to automatically close issues

Helpful resources

Step 9: Merge Pull Request

Click

Merge Pull Request

Then

Confirm Merge

Screenshot 9: Successfully merged PR.

The screenshot displays the GitHub interface during a pull request merge. At the top, a comment from user Sriarevalagan23, posted 1 minute ago, states: "new changes have been made in the read me file." Below this, a commit history entry shows a commit titled "Updated README with project description" with commit hash 5e58284. The main part of the interface is a "Commit message" dialog box. The "Commit message" field contains the text "Merge pull request #1 from Sriarevalagan23/update-readme". The "Extended description" field contains the text "Updated README with project description". At the bottom of the dialog, it states "This commit will be authored by 158122890+Sriarevalagan23@users.noreply.github.com." and provides two buttons: "Confirm merge" (in green) and "Cancel". Below the dialog box, a purple notification banner with a GitHub logo icon reads: "Pull request successfully merged and closed". The text continues: "You're all set — the `update-readme` branch can be safely deleted. If you wish, you can also delete this fork of Santhosh-1307/Test-repo in the [settings](#)." A "Delete branch" button is located on the right side of the notification banner.

Step 10: Pull Request Link

Example:

<https://github.com/Santhosh-1307/Test-repo/pull/1>

Summary

- Forked a public GitHub repository.
- Cloned it to the local system.
- Created a new feature branch.
- Modified the README file.
- Committed and pushed the changes.
- Created a Pull Request.
- Reviewed the Pull Request and addressed feedback.
- Successfully merged the Pull Request into the main branch.

Result

The fork-and-pull request workflow was successfully implemented using Git and GitHub, demonstrating collaborative software development.